Google Cloud H2S

AI Builder Cup

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Team Details

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Problem Statement: Automated Multimodal Motor Insurance Claim Fraud Detection & Triaging

About the idea

Motor Claims Fraud Detector

A multimodal AI system that automates motor insurance claim fraud detection. It ingests complete claim packs -- PDF claim forms, repair quotations, police/survey reports, and vehicle damage photographs -- and evaluates eight specialized fraud detectors (D1-D8) to produce an adjuster-ready investigation report with transparent, evidence-backed reasoning.

The Problem

Motor insurance fraud costs the industry billions annually. Claims adjusters face high caseloads, inconsistent evaluation, and difficulty cross-referencing photos, documents, and pricing data simultaneously. Manual review is slow, subjective, and cannot scale.

Our Approach

We reduce toil and improve fraud detection. We use Google Gemini multimodal capabilities in a two-stage pipeline: Stage 1 (Vision) classifies every photograph into damage classes (crack, dent, scratch, glass shatter, lamp broken, tire flat), then Stage 2 (Text) synthesizes documents, photo findings, price books, and prior claims history against eight rule-based fraud detectors. The system outputs a structured JSON verdict with fraud probability, confidence, evidence chains, and recommended next steps.

Opportunities

How is it different from existing solutions?

•Most fraud detection tools are rules-only or metadata-only -- they cannot read photographs. Our system uses Gemini native multimodal vision to classify physical damage directly from images.

•We cross-reference photos against documents in a single pipeline -- detecting eight specialized fraud types in one pass.

•The system is transparent: every verdict includes named evidence (document line or photo file), not opaque scores. Adjusters can audit every decision.

How does it solve the problem?

•Reduces claim review time from hours to minutes (2-5 min per claim pack).

•**Automated Triaging:** Separates clean claims from suspicious packs with high fraud probability (0–100%). Provides specific recommended actions: 'Pay Claim', 'Audit Workshop', or 'Refer to SIU'.

•**Evidence Calibration:** Cross-references repair estimates against an authoritative market price book and prior loss histories.

•Reduces human inconsistency across eight standardized fraud detectors.

USP: Unique Selling Proposition

End-to-end multimodal fraud detection -- from raw photographs to verdict -- powered by Gemini vision and language models, with serverless execution on Google Cloud, and full evidence traceability.

Core Features

Two-Stage Multimodal AI Pipeline

Stage 1: Gemini Vision classifies damage photos. Stage 2: Gemini Text synthesizes documents + photo findings against D1-D8 detectors.

Comprehensive Adjuster Reports

Covers Eight Specialized Fraud Detectors (D1-D8) - Ghost damage, Severity inflation, Pre-existing/staged damage, Price/invoice inflation, Policy exclusion gate, Duplicate claims, Wear-and-tear, Loss-type mismatch. Generates summary verdicts, damage ledgers, red flags, confidence scores, and downloadable markdown reports.

SHA-256 Photo Deduplication and Caching

Cryptographic hashing detects identical photos across claims. Cached results avoid redundant API calls. Cross-claim duplicate fraud detection built-in.

Cloud-Native Serverless Architecture

Cloud Run (UI) > Cloud Tasks (rate-limited queue) > Cloud Run Jobs (analysis). Auto-scaling, scales-to-zero, exponential backoff retries.

GCS Lifecycle Cost Optimization

Automatic tiering: Standard > Nearline (30d, 70% savings) > Coldline (90d, 85% savings) for audit compliance.

Automatic Multi-Model Failover Chains

Vision: gemini-3.8-flash > 3.7-flash > 3.6-flash. Text: gemini-3.7-flash > 3.5-flash-lite > 2.5-flash. Automatic quota failover on HTTP 429.

Live Web UI with Drag and Drop

Real-time progress tracking. Markdown report export. Deployed on Cloud Run (asia-southeast1).

Process flow diagram

1. Claim Upload

Adjuster uploads claim form PDF, repair estimate, & photos via browser UI.

2. Cloud Ingestion

Files saved to GCS bucket; tasks enqueued to Cloud Tasks.

3. Stage 1: Vision AI

Gemini classifies each photo for damage.

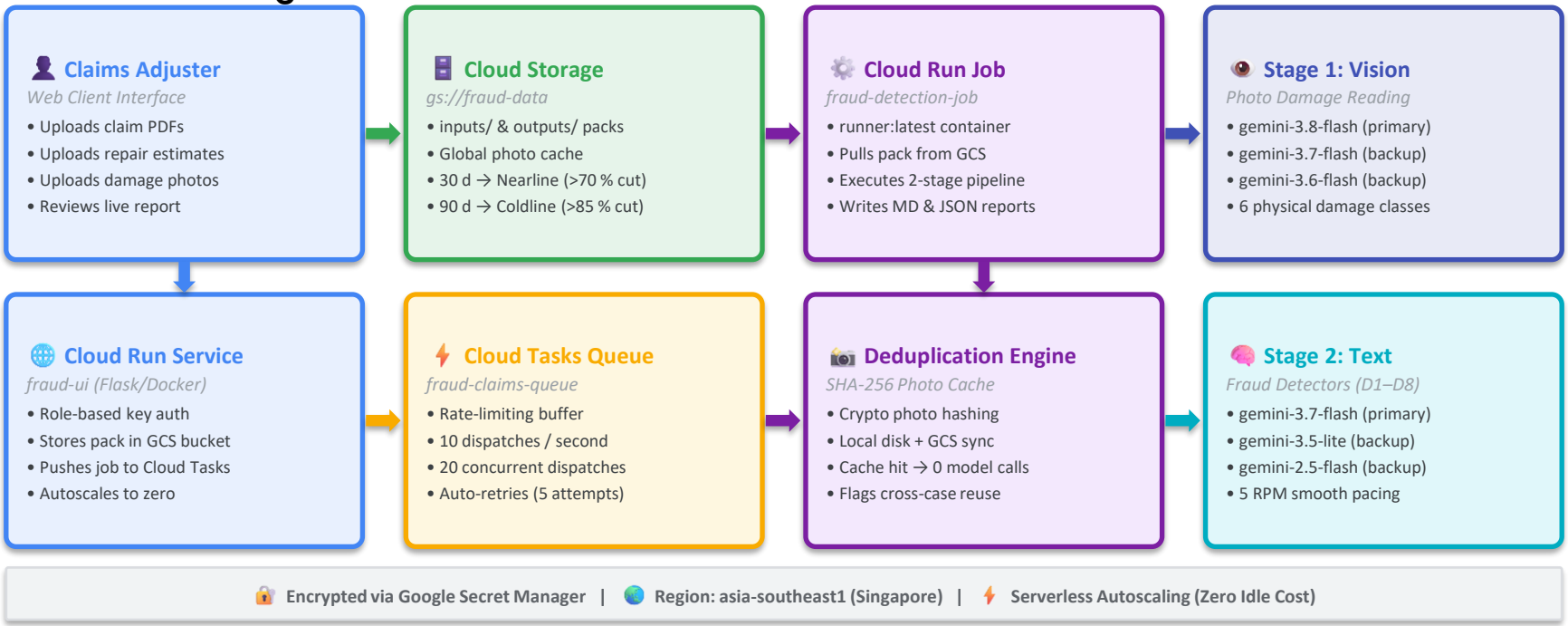
4. Stage 2: Fraud Eval

Text model synthesizes quotation, narrative, policy dates, and photo readings against D1–D8 fraud detectors.

5. Report Generation & Triage

Verdict (clean/fraud), probability, and adjuster action rendered live & downloadable.

Architecture diagram



Tech Stack

- **AI / Foundation Models:** Google Gemini 3.8 Flash, 3.7 Flash, 3.6 Flash, 3.5 Flash Lite, 2.5 Flash. Also OpenAI-compatible API endpoint (supports llama.cpp, vLLM, Ollama, LM Studio)
- **Cloud Infrastructure:** Google Cloud Platform (Cloud Run, Cloud Tasks, Cloud Storage, Secret Manager, Artifact Registry, Cloud Build)
- **Backend Runtime:** Python 3.12, Flask, Gunicorn, pypdf (document text extraction), Pillow (image handling)
- **Frontend Interface:** Modern HTML5, Vanilla CSS3 (Dark mode, Glassmorphism design tokens), Responsive Vanilla JS
- **Caching & Deduplication:** SHA-256 cryptographic hashing with hybrid local and GCS persistent storage
- **Containerization:** Docker, Google Artifact Registry (asia-southeast1-docker.pkg.dev)

Estimated implementation cost

Cost Structure: Designed for Free-Tier / Minimal Spend

The entire prototype runs within Google Cloud free tier and Gemini API free quotas. Production scaling costs are outlined below.

Current Prototype (Free Tier)

- Cloud Run: Free tier (2M requests/month, 360K vCPU-seconds)
- Cloud Storage: Free tier (5 GB Standard, auto-tiering to Nearline/Coldline)
- Cloud Tasks: Free tier (1M operations/month)
- Gemini API: Free tier with rate pacing (5 RPM, 12s intervals)
- Estimated monthly cost: $0 (within free tier limits)

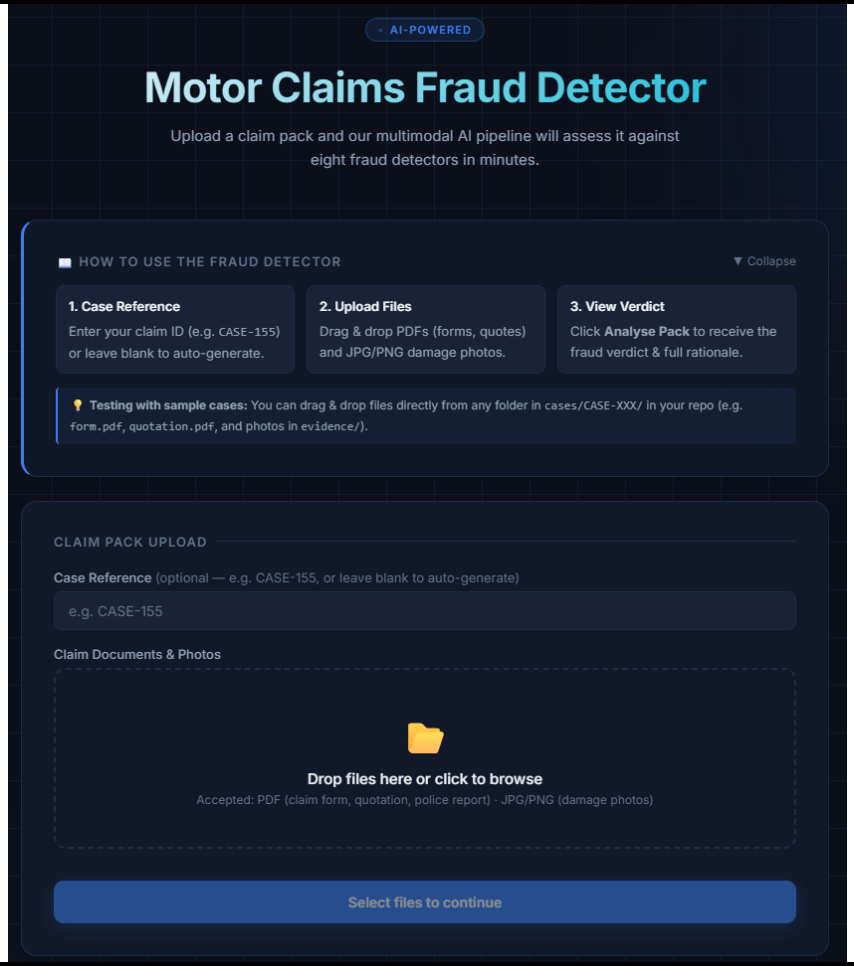
Enterprise Production Scale

- Vertex AI: Pay-as-you-go Gemini with 1,000-10,000+ RPM ($0.075/1K input tokens)
- Cloud Run: ~$50-200/month for sustained concurrency
- Cloud Storage: ~$10-50/month for 100 GB+ with lifecycle optimization
- Gemini Batch API: 50% discount for non-urgent bulk rescoring
- Estimated production cost: $200-500/month for ~1,000 claims/month

Prototype Snapshots

Live Prototype Highlights

- **Clean Modern Interface:** Responsive web dashboard accessible from desktop or mobile devices.
- **Secure Authorization:** Restricted access field protected by Google Secret Manager (turned off).
- **Live Progress Tracker:** Real-time visual feedback on document upload, queueing, and model execution.



Prototype Snapshots

Motor Claims Fraud Detection Report

Generated: 2026-10-18 08:26:54 | **Duration:** 1.1 min

Summary

Case	Verdict	Fraud Probability	Confidence	Primary Detector	Finding	Recommended Action
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CASE-635	**fraud**	85%	high	D3	Claimant asserts a high-speed rear impact by a van destroying the rear, door, and boot li...	escalate to SIU

Detailed Case Analysis

CASE-635 -- FRAUD (85%)

`##################...` **85%** likelihood of fraud | confidence: **high**

Primary finding (D3 / Pre-existing or staged damage / Narrative-evidence contradiction): Claimant asserts a high-speed rear impact by a van destroying the rear, door, and boot lid rendering the vehicle a total loss, but photographic evidence shows only cosmetic scratches and surface spiderweb cracking.

Recommended action: `escalate to SIU`

Live Prototype Highlights

- **Comprehensive Output:** Verdict summary table, photographic evidence ledger, triggered fraud rules, and markdown export.
- **Tested with Sample Cases**: Validated on 13 claim packs in Singapore motor insurance format.

Prototype Performance report

Measured Performance & Reliability Metrics

- **Evaluation Dataset:** 13 anonymized motor claim packs (5 clean + 8 fraudulent), covering all 8 detector types (D1-D8) across easy, medium, and hard difficulty levels.
- **Headline Fraud Detection Accuracy:** 10/13 cases correctly classified (77% accuracy), with 100% detection on severe fraud.*
- **Zero False Negatives:** All fraudulent cases were successfully caught — zero fraudulent claims were mistakenly passed as clean.
- **Photo Deduplication Speedup:** Stage 1 photo reading dropped from ~15 seconds to <0.1 seconds on cached images.
- **Zero Quota Drops:** Cloud Tasks rate limiting + 5 RPM pacing achieved 100% request success without HTTP 429 quota exhaustion.
- **Storage Cost Reduction:** Automated GCS lifecycle rules reduce long-term compliance storage costs by over 70% to 85%.

* on small quantized models

Future Development Roadmap

> Confidence Calibration and Human Feedback Loop

Track adjuster override rates to calibrate confidence thresholds and improve detector precision over time.

> Syndicated Fraud Ring Detection

Connect to BigQuery data lakes and Vertex AI Vector Search to identify cross-insurer fraud syndicates, repeat claimants, and staged accident rings across millions of historical claims.

> Gemini Batch API Integration

Use Gemini Batch API for non-urgent claims to achieve 50% cost reduction on bulk rescoring.

> Archive Storage Tier

Add Archive class (365+ days, ~$0.0012/GB/mo, 95%+ savings) for multi-year regulatory audit retention.

Project Links

- **GitHub Public Repository:** https://github.com/kevinseow/motor-claims-fraud-detection
  Contains complete source code, tests, documentation, and deployment configs
- **Demo Video Link: https://youtu.be/Y5a8I_BQPuQ**
- **Final Product Link:** https://fraud-ui-1082243885044.asia-southeast1.run.app

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Thank you

